SIMATS ENGINEERING

ASSESSMENT TOOL CO4- SDG

COURSE CODE: ITA 0605 **COURSE NAME: MACHINE LEARNING**

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Unit IV –INSTANT BASED LEARNING (K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning) from your Machine Learning syllabus,

I have prepared 40 analytical assignment questions aligned with relevant

UN Sustainable Development Goals (SDGs) wherever applicable. The assignments emphasize data analysis, critical thinking, and real-world problem solving, matching the course outcomes.

QUESTION

2. Use KNN to predict diabetes risk based on patient health records and evaluate its performance.

KNN-Based Diabetes Risk Prediction Using Patient Health Records

1. Title

Use K-Nearest Neighbors (KNN) to predict diabetes risk based on patient health records and evaluate its performance.

2. Aim

To develop a KNN classification model that predicts whether a patient is likely to have diabetes based on health-related attributes and to evaluate the model using suitable performance metrics.

3. Objective

- To analyze patient health records related to diabetes.
- To preprocess the dataset for machine learning.
- To apply the K-Nearest Neighbors classification algorithm.
- To classify patients into diabetic and non-diabetic categories.
- To evaluate the prediction performance using a confusion matrix, accuracy, precision, recall, and F1-score.

4. Dataset

The model can be developed using the Pima Indians Diabetes Dataset. The dataset contains patient health records along with a diabetes outcome.

Important attributes include:

- Pregnancies – Number of pregnancies.
- Glucose – Plasma glucose concentration.
- Blood Pressure – Diastolic blood pressure.
- Skin Thickness – Skin fold thickness.
- Insulin – Serum insulin level.
- BMI – Body Mass Index.
- Diabetes Pedigree Function – Family history-related diabetes risk.
- Age – Age of the patient.
- Outcome – Target variable indicating whether diabetes is present.

Outcome:

- 0 – Non-diabetic
- 1 – Diabetic

5. Data Preprocessing

Before applying KNN, the dataset must be prepared properly.

The preprocessing steps are:

1. Check the dataset for missing or invalid values.
2. Handle zero or missing values in health attributes where appropriate.
3. Separate input features from the target variable.
4. Divide the dataset into training and testing sets.
5. Normalize the numerical features because KNN is distance-based.

Feature scaling is particularly important because attributes such as glucose, age, BMI, and insulin have different numerical ranges.

6. Working of KNN

KNN classifies a new patient by comparing the patient's health record with existing training records.

The basic process is:

1. Select a value for K.
2. Calculate the distance between the new patient and training patients.
3. Identify the K nearest patients.
4. Examine their diabetes classifications.
5. Assign the class that occurs most frequently among the nearest neighbors.

For example, if K = 5 and three of the five nearest patients are diabetic while two are non-diabetic, the new patient is classified as diabetic.

7. Distance Calculation

KNN commonly uses Euclidean distance to measure similarity between patients.

Euclidean distance can be represented as:

Distance = $\sqrt{[(x_1-y_1)^2 + (x_2-y_2)^2 + ... + (x_n-y_n)^2]}$

Here, x represents the health attributes of the new patient and y represents the attributes of a training patient.

A smaller distance indicates that two patient records are more similar.

8. Selection of K

The value of K affects the performance of the model.

- Small K values make the model sensitive to individual records and noise.
- Large K values make the model more stable but may reduce its ability to identify local patterns.
- Different K values can be tested to determine the value that provides the best validation performance.

A suitable K value can be selected by comparing model performance for several K values.

9. Prediction Process

For a new patient, the model takes health information such as glucose level, BMI, age, blood pressure, insulin, and other available attributes.

The model then:

Patient Health Record → Feature Scaling → Distance Calculation → K Nearest Patients → Majority Voting → Diabetes Prediction

The final prediction is either:

- Diabetic
- Non-diabetic

10. Confusion Matrix

The confusion matrix is used to understand the types of predictions made by the model.

It contains four categories:

- True Positive (TP): Patient has diabetes and the model predicts diabetes.
- True Negative (TN): Patient does not have diabetes and the model predicts non-diabetic.
- False Positive (FP): Patient does not have diabetes but the model predicts diabetes.
- False Negative (FN): Patient has diabetes but the model predicts non-diabetic.

A confusion matrix helps identify whether the model is making more false-positive or false-negative predictions.

11. Performance Evaluation

The KNN model can be evaluated using the following metrics.

Accuracy:

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Accuracy represents the overall percentage of correctly classified patients.

Precision:

Precision = TP / (TP + FP)

Precision indicates how many patients predicted as diabetic were actually diabetic.

Recall:

Recall = TP / (TP + FN)

Recall indicates how effectively the model identifies patients who actually have diabetes.

F1-Score:

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

F1-score provides a balance between precision and recall.

12. Example Evaluation

Assume the model produces the following results on the test dataset:

- True Positive = 40
- True Negative = 85
- False Positive = 15
- False Negative = 20

Total predictions = 160.

Accuracy:

Accuracy = (40 + 85) / 160 = 78.13%

Precision:

Precision = 40 / (40 + 15) = 72.73%

Recall:

Recall = 40 / (40 + 20) = 66.67%

F1-Score:

F1-Score ≈ 69.57%

These values indicate that the model correctly classifies a significant portion of the test patients, while there is still room for improvement in identifying diabetic patients.

13. Interpretation of Results

The KNN model uses similarities between patient health records to make predictions. A higher accuracy indicates better overall classification performance.

For diabetes prediction, recall is particularly important because a false negative means that a patient who may have diabetes is predicted as non-diabetic. Therefore, the model should not be evaluated using accuracy alone.

A balanced model should provide good accuracy while maintaining suitable precision and recall.

14. Advantages of KNN

- Simple and easy to understand.
- Requires no complex training process.
- Works well with numerical health records.
- Can classify new patients based on similar existing records.
- Useful for small and medium-sized datasets.

15. Limitations

- Prediction can become slower for large datasets.
- Performance depends strongly on the selected K value.
- Feature scaling is important.
- Irrelevant or noisy features can affect predictions.
- The model can be sensitive to the distribution of the training data.

16. Conclusion

K-Nearest Neighbors can be used to predict diabetes risk from patient health records by identifying patients with similar health characteristics. After preprocessing and feature scaling, KNN classifies new patients using the majority class among their nearest neighbors. The model can be evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. The results help determine how effectively KNN can distinguish between diabetic and non-diabetic patients.